INDUSTRIAL VISIT REPORT

AIRAA ENGINEERING, COIMBATORE

Host Company	Airaa Engineering, Coimbatore, Tamil Nadu
Date of Visit	14 August 2026 (Friday)
Department	Mechanical Engineering
Focus Areas	Fiber laser cutting, MIG welding, sheet metal fabrication

1. Introduction

As part of our course work, our class went to Airaa Engineering in Coimbatore for an industrial visit. It's a sheet-metal fabrication unit, and the idea was to see how the things we've been reading about in Manufacturing Processes, Materials, and CAD/CAM classes actually happen on a shop floor.

The main things we got to see were fiber laser cutting and MIG welding. We watched a plain sheet of metal get turned into a finished, cut-to-shape component right in front of us, and the staff there walked us through the materials, gases, and software that go into it.

2. About Airaa Engineering

Airaa Engineering is a fabrication and job-work unit based in Coimbatore. They mainly do CNC fiber laser cutting and MIG welding for other companies — basically, customers send drawings and they cut the components to size in mild steel, stainless steel, GI, and aluminium. We couldn't find much about the company online beyond this, so everything here is based on what we were shown and told on-site.

3. Objectives of the Visit

- Understand how fiber laser cutting actually works, not just in theory.
- See MIG welding being done and how it's used in fabrication work.
- Learn about the different sheet materials — MS, GI, SS, aluminium — and how they're processed.
- Find out why different gases (oxygen and nitrogen) are used for different metals.
- Understand the difference between hot-rolled and cold-rolled mild steel.
- Get a look at the CNC control panel and the cutting software.
- Understand why a stable power supply matters so much for a machine like this.
- See how a CAD drawing eventually becomes a cutting program on the machine.

4. Laser Cutting Process

This was the main thing we went to see. Airaa Engineering runs a 3 kW fiber laser cutting machine that cuts flat sheet metal into shapes based on customer drawings. LASER, as our guide reminded us, stands for Light Amplification by Stimulated Emission of Radiation — the machine focuses a very intense beam onto the sheet, which melts and burns through the metal along the programmed path, giving a clean, narrow cut.

The machine has a CNC control unit with a touchscreen, a keyboard, and a big red emergency-stop button (you can see it in Fig. 1 and Fig. 3). From there the operator loads the cutting file, watches the cut happen live, and can tweak the speed or power if needed.

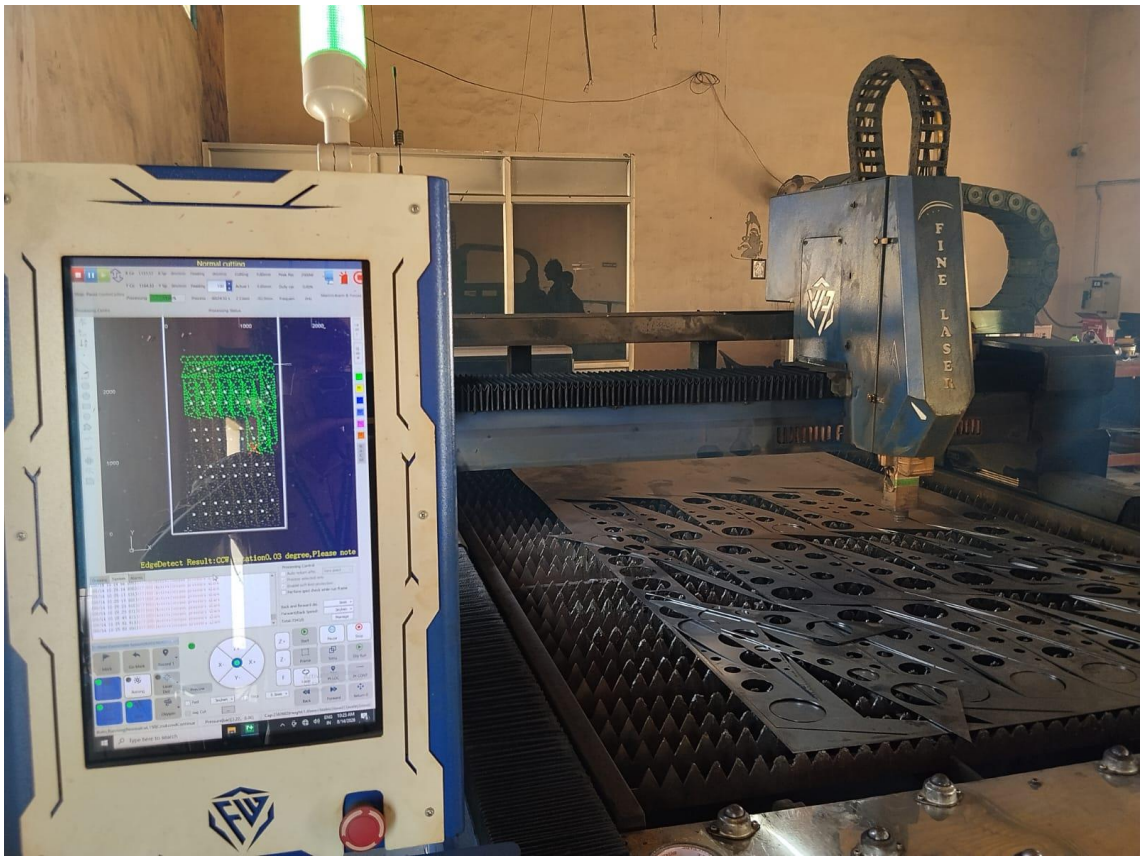


Fig. 1 The laser cutting machine mid-operation — control screen on the left, cutting bed with finished sheet on the right

5. Materials Handled

A few different sheet materials are cut at the unit, depending on what the customer needs:

- Mild Steel (MS) — comes as either Hot Rolled (HR) or Cold Rolled (CR) sheet.
- Galvanized Iron (GI) sheet.
- Stainless Steel (SS).
- Aluminium / aluminized sheet.

One thing that was explained clearly is the difference between HR and CR mild steel. HR sheet is rolled while the steel is still hot and then shaped, so it comes out a bit rough and is cheaper. CR sheet is rolled cold, without reheating, so the finish is smoother and the dimensions are more accurate — it costs a little more because of that extra precision.

Roughly, the thickness ranges they work with are:

Material	Thickness Range	Remarks
Mild Steel (MS)	0.5 – 20 mm	Cut with Oxygen (O_2)
Stainless Steel (SS)	1 – 10 mm	Cut with Nitrogen (N_2)
GI / Aluminium	1 – 3 mm	Cut with Nitrogen (N_2)

6. Gases Used in Cutting

Something we didn't fully appreciate before the visit is how much the choice of gas matters. Two gases are used, and each one is picked for a reason:

- Oxygen (O_2) — used for mild steel. It reacts with the hot metal and actually helps burn through it, which is useful for thicker MS sheets.
- Nitrogen (N_2) — used for stainless steel, GI, and aluminium. It doesn't react with the metal, so it keeps the cut edge clean and prevents oxidation/discoloration.

The pressures used were also mentioned to us:

Gas	Pressure	Notes
Nitrogen (N_2)	~14 bar	Stored as liquid nitrogen; pressure builds up in the tank before use
Oxygen (O_2)	~6 bar	Stored as liquid O_2 / in cylinders of about 230 litres

Both gases are kept in liquid form on site, and the pressure is allowed to build up before it's fed to the cutting head — this keeps the gas flow steady during the actual cut.



Fig. 2 Our group with the plant supervisor on the shop floor — laser cutting machine and cut sheet stock visible behind us

7. Software and CNC Programming

The cutting machine runs on software called Ray2S — the operator said it's fairly simple to use once you know it. You load in the design, arrange the parts on the sheet so there's as little wastage as possible (called nesting), and set your cutting speed, power, and gas pressure from there.

Before that stage, the actual design is done in AutoCAD and saved out as a DWG or DXF file. That DXF file is what gets imported into Ray2S, which turns the drawing into G-code — the actual set of instructions the machine follows to move and cut. So the whole path is: draw it in AutoCAD → export as DXF → import into the cutting software → machine cuts it out.

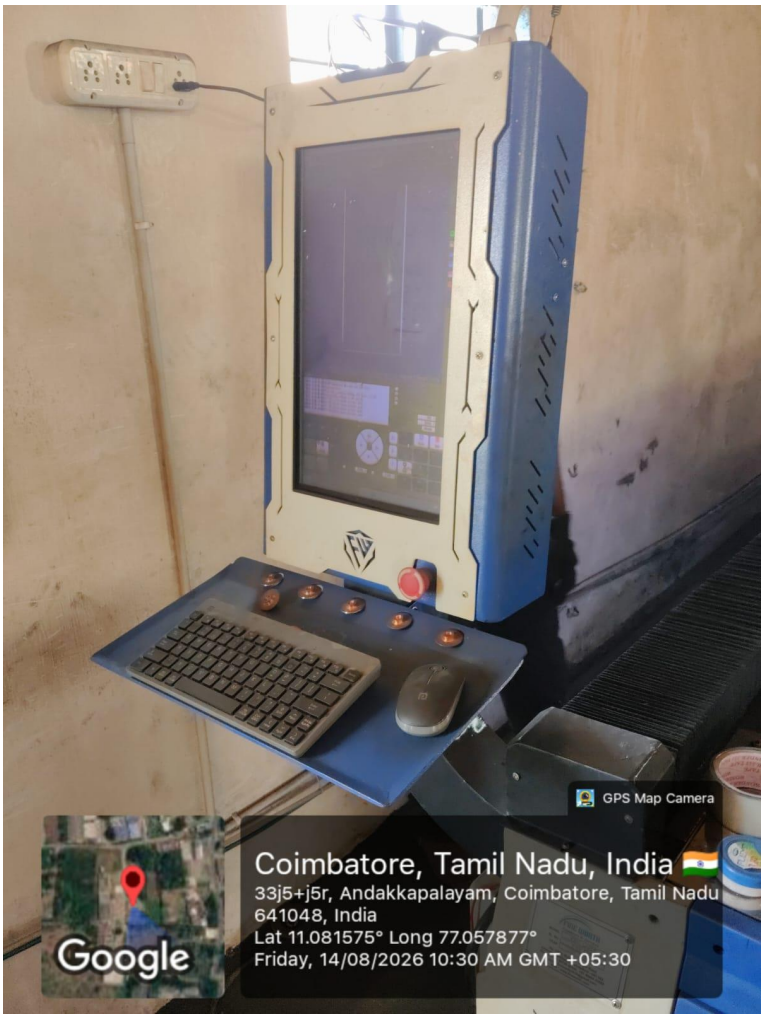


Fig. 3 Control panel of the laser cutting machine — touchscreen, keyboard, and control buttons

8. Power Supply

Since even a tiny power dip mid-cut can ruin a job, the unit runs the machine through an online UPS connected via an MCB panel. The UPS is backed by a battery, so it keeps supplying steady power (around 13–14 volts at the control side) even if there's a fluctuation from the main line. There's also a chiller running alongside the machine to keep the laser source and optics cool, since they heat up during continuous cutting.

9. MIG Welding

Besides laser cutting, the unit also does MIG welding to join parts after they're cut and fabricated. In MIG welding, a wire electrode is fed continuously through the welding gun while a shielding gas protects the weld from air contamination. It's generally faster and easier to work with than manual arc welding, which is probably why it's the go-to method here for mild steel joints.

10. Quality Check

One small but interesting detail — after the sheet is placed on the bed, the machine runs an edge-detection check to see if the sheet is slightly misaligned (they mentioned something like a 0.03-degree rotation error). It corrects for that automatically before starting the actual cut. It's a good reminder of why dimensional tolerance and inspection matter so much in manufacturing — even something this small can throw off a whole batch of parts if it's ignored.

11. What We Learned

Overall, this visit gave us a much clearer picture of:

- How fiber laser cutting works and where it's used in industry.
- Why oxygen and nitrogen are used differently depending on the metal.
- The practical difference between hot-rolled and cold-rolled mild steel.
- Roughly what thickness ranges MS, SS, GI, and aluminium sheets are cut in.
- How a design moves from AutoCAD to a finished part — DXF, Ray2S, G-code, and the machine itself.
- Why a reliable, UPS-backed power supply is a big deal for precision machines.
- The basics of how MIG welding is used alongside cutting in fabrication.

12. Conclusion

Overall, the visit to Airaa Engineering was genuinely useful — a lot more useful than just reading about laser cutting in a textbook. Seeing the machine actually running, hearing why certain gases and materials are chosen, and watching the whole process from a CAD file to a finished cut sheet made a lot of the theory click into place.

It also gave us a better sense of how much precision, the right material choice, and something as basic as a stable power supply all matter in real manufacturing. It was a good, practical addition to what we've been learning in class, and it's given us a better feel for what this side of Mechanical Engineering actually looks like on the ground.